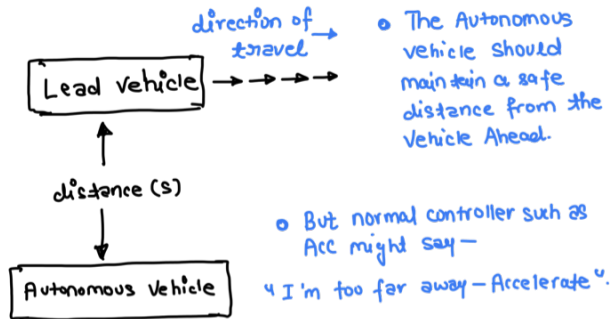


Can Control Barrier func. keep Automated Vehicles Safe in Live freeway traffic?

If an Autonomous vehicle is driving in real traffic, can a Control Barrier fn. (CBF), act as a safety layer & prevent the vehicle from violating a desired safety condition?

Authors put a CBF supervised automated vehicle into real freeway traffic, near Nashville Tennessee, and collected real driving data.



What exactly is a CBF?

A control barrier fn. defines a mathematical quantity $h(x)$

whether the vehicle is inside a given safe region. e.g. $h(x) \geq 0$ (Safe)

$h(x) < 0$; safety-property violated.

paper describes: $S = \{x : h(x) \geq 0\}$

4 The car should probably keep some dist. These states are safe, these are Unsafe.

They used a time-gap safety property.

$$h = s - (t_{\min} v + d_{\min})$$

where:

s = dist / gap between vehicles

v = autonomous vehicle speed

$t_{\min}$ = min. desired time-gap

$d_{\min}$ = min. distance offset.

This means the required safe distance depends on how fast the autonomous vehicle is travelling.

Example to understand 'h'.

suppose, $v = 20 \text{ m/s}$

Then the required safe gap is:

$$s_{\min} = 2(20) + 15$$

$$s_{\min} = 55 \text{ m}$$

So:

Case I → Actual gap = 70m

Then $h = 70 - 55 \text{ m} \Rightarrow 15 \text{ m}$

Therefore $h > 0$

Case II → $s = 55 \text{ m} \rightarrow h = 0$

exactly the safe boundary

Case III → $s = 45 \text{ m}$ (Actual gap)

$$h = -10 < 0$$

The vehicle has violated the defined time-gap safety property

But How does CBF actually control the car?

The vehicle model used in the paper is:

$$\dot{p} = v \quad \dot{v} = u \quad \text{where:}$$

p = vehicle position ; v = vehicle velocity
 u = command acceleration.

So, they assume acceleration is the control

$$\text{input: } E = \begin{bmatrix} s \\ \Delta v \end{bmatrix} \quad \Delta v = v_{\text{lead}} - v_{\Delta v}$$

Another important eqn. in paper $\Rightarrow$

$$\text{we have: } h = s - (t_{\min} v + d_{\min})$$

differentiate

$$\dot{h} = \dot{s} - t_{\min} \dot{v}$$

relative speed is: $\dot{s} = \Delta v$ & $\dot{v} = u$

$$\therefore \dot{h} = \Delta v - t_{\min} u$$

What does this eqn. mean physically?

Suppose the follower is getting too close:

Then, $h \rightarrow 0$

and potentially $\dot{h} < 0$

We don't want 'h' to keep decreasing

The CBF imposes: $\dot{h} + k h > 0$ where $k > 0$

CBF acceleration constraint -

substitute, $\dot{h} = \Delta v - t_{\min} u$

$$\text{into } \dot{h} + k h \geq 0$$

we get, $\Delta v - t_{\min} u + k h \geq 0$

rearrange, $t_{\min} u \leq k h + \Delta v$

$$\therefore u \leq \frac{k h + \Delta v}{t_{\min}}$$

The paper writes this as:

$$u_{\text{CBF}} \leq \frac{1}{t_{\min}} [k(s - (t_{\min} v + d_{\min})) + \Delta v]$$